StudyOpace

Computer Network

UNIT - 1











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1.1 INTRODUCTION TO COMPUTER NETWORKS

DEFINITION OF COMPUTER NETWORK

- A **computer network** is a collection of two or more computers (or devices) that are connected together using a communication medium (wired or wireless) to share data, resources, and information.
- In simple words, a computer network is a system where computers "talk" to each other and exchange information.
- The connection can be through cables (wires), optical fibers, or wireless signals (radio waves, satellites, etc.).
- Example: When you connect your laptop to Wi-Fi and browse the internet, you are using a computer network.

Why Do We Need Computer Networks? (Purpose)

- To **share data** between multiple users quickly and easily, instead of manually copying files using pen drives.
- To **share hardware resources** like printers, scanners, and storage devices among many computers, saving cost.
- To **enable communication** between people through email, video calls, and messaging.
- To allow **centralized data storage**, so that important data can be stored at one place (server) and accessed by many users.
- To provide **remote access**, so that a person can access files or systems from a different location.
- **Exam Tip:** If asked "Why are computer networks needed?", write these points with small explanations — this is a common 2-mark or sub-question.

Basic Components of a Computer Network

- **Sender** – the device that sends the data (example: your computer).
- **Receiver** – the device that receives the data (example: server or another computer).
- **Medium** – the physical path through which data travels (cable, fiber, or air for wireless).
- **Message** – the actual data being transferred (text, image, video, etc.).
- **Protocol** – a set of rules that both sender and receiver agree to follow for successful communication.

1.1.4 How Data Travels in a Network (Basic Working)

- The sender converts the data into a suitable format (called signals) for transmission.
- The data is broken down into smaller pieces (called **packets**) before sending, so that transmission is efficient and errors can be managed easily.

- These packets travel through the transmission medium (wire/wireless) following certain rules (protocols).
- The receiver collects all packets, reassembles them in the correct order, and converts them back into readable data.
- **Example:** Sending a WhatsApp message breaks the message into packets, sends them over the mobile network, and the receiver's phone reassembles them into the original message.

1.2 USES OF COMPUTER NETWORKS

Computer networks are used in almost every field today. SPPU syllabus specifically asks about **three types of uses**: Business Applications, Home Applications, and Mobile Users.

1.2.1 BUSINESS APPLICATIONS

- **Purpose:** Networks allow companies/organizations to share resources, information, and communicate efficiently, which increases productivity and reduces cost.
- **Resource Sharing**
  - Multiple employees can share the same printer, scanner, or storage device connected on the network, instead of buying a separate device for every employee.
  - This reduces hardware cost significantly for companies.
- **Client-Server Model in Business**
  - Companies store important data (like employee records, customer data) on a powerful central computer called a **server**.
  - Employees use their computers (called **clients**) to request and access this data from the server whenever needed.
  - This is known as the **Client-Server Model**, and it is one of the most important architecture models used in business networks.
- **Communication Medium for Business**
  - Networks allow employees to communicate using **email**, video conferencing, and instant messaging, saving time and travel cost.
  - Businesses can conduct **video meetings** with clients located in other cities or countries.
- **E-commerce (Doing Business Electronically)**
  - Networks enable **online business transactions**, such as online shopping, online banking, and online bill payment.
  - Example: Amazon, Flipkart use computer networks to let customers order products remotely.

- **VPN (Virtual Private Network) for Business Security**
  - Businesses use **VPN** to allow employees to securely connect to the company's private network from remote locations (like home) over the public internet.
  - This ensures that company data remains secure even when accessed from outside the office.
- **Advantages of Business Applications of Networks:**
  1. Reduces operational cost by sharing hardware and software resources.
  2. Increases employee productivity through fast communication.
  3. Enables remote working and global business operations.
  4. Provides centralized data management and better data security through servers.
- **Applications:** Banking systems, online shopping, video conferencing tools (Zoom, Teams), company intranets, cloud-based business tools.

1.2.2 HOME APPLICATIONS

- **Purpose:** Networks are used at home for entertainment, communication, and easy access to information.
- **Access to Remote Information**
  - Home users can access the **internet** to browse news, government websites, online shopping, and general information from anywhere.
  - This includes accessing **online newspapers**, **online magazines**, and **digital libraries**.
- **Person-to-Person Communication**
  - Networks allow home users to send **emails**, use **social media** (Facebook, Instagram), and make **video calls** (WhatsApp, Skype) with family and friends.
  - This has replaced traditional methods like letters and telegrams.
- **Interactive Entertainment**
  - Home networks are used for **online gaming**, streaming movies and music (Netflix, YouTube, Spotify), and interactive TV.
  - Multiplayer online games use networks to connect players from different locations in real time.
- **Electronic Commerce (E-Commerce) at Home**
  - Home users use networks to do **online shopping**, pay bills, book tickets, and do **online banking** directly from their homes.

1.2.3 MOBILE USERS

- **Purpose:** Mobile networks allow users to stay connected to the network while moving from one place to another, without needing a fixed wired connection.
- **What Makes Mobile Networking Special**
  - Unlike traditional wired networks, **mobile users** are not fixed at one location; they can move while still being connected.
  - Mobile devices use **wireless communication** technologies like 4G, 5G, and Wi-Fi to stay connected.
- **Applications for Mobile Users**
  - **Text messaging (SMS)** and internet-based messaging apps (WhatsApp).
  - **Mobile banking and mobile commerce** – allowing users to do transactions using smartphones.
  - **GPS-based Navigation** – networks help in location tracking and route navigation (Google Maps).
  - **Mobile Internet Access** – checking emails, browsing social media, and video calls while traveling.
- **Challenges in Mobile Networking**
  - Mobile signals can face **interference** from buildings, weather, or distance from towers.
  - **Handover/Handoff** is needed — when a mobile user moves from one network tower's area to another, the connection must be smoothly transferred without call/data drop.
- **Advantages of Mobile Networking for Users:**
  1. Provides connectivity anytime and anywhere without wires.
  2. Supports real-time navigation and location-based services.
  3. Enables mobile banking and mobile commerce.
  4. Increases convenience for users who travel frequently.
- **Applications:** Smartphones, GPS navigation systems, mobile banking apps, IoT wearable devices.

1.3 NETWORK HARDWARE

- **Definition:** Network hardware refers to the physical devices and classification of networks based on their size, coverage area, and the number of devices they connect. SPPU syllabus specifically asks about **PAN, LAN, MAN, WAN, and Internetworks**.
- **Classification Basis:** Networks are mainly classified based on **transmission technology** (Broadcast vs Point-to-Point) and **scale/size** (PAN, LAN, MAN, WAN).

1.3.1 TRANSMISSION TECHNOLOGY (BASIS OF CLASSIFICATION)

- **Broadcast Networks**
  - In broadcast networks, a **single communication channel is shared** by all the devices (machines) in the network.
  - Any message sent by one device is received by **all other devices** on the network.
  - Every device checks the **destination address** in the message; if the address matches its own, it processes the message, otherwise it ignores it.
  - Broadcast networks usually allow **addressing a packet to all destinations** (called **broadcasting**) or to a **subset of machines** (called **multicasting**).
  - Example: Traditional Ethernet LAN, wireless networks.
- **Point-to-Point Networks**
  - In point-to-point networks, there are **many individual connections** between pairs of machines.
  - To go from source to destination, a packet may have to pass through **several intermediate machines/routers** before reaching its destination.
  - Multiple routes of different lengths may exist, so **routing algorithms** play an important role in point-to-point networks.
  - Example: WAN connections, Wide Area Networks using routers.
- **Exam Tip:** This classification (Broadcast vs Point-to-Point) is frequently confused with LAN vs WAN — remember, this is based on **how data is transmitted**, not on network size.

1.3.2 PAN (PERSONAL AREA NETWORK)

- **Definition:** A **Personal Area Network (PAN)** is a network used for communication between devices (like computer, mobile phone, tablet) within a very short range, usually around a **single person**.
- **Range:** Typically covers a range of a few meters (1 to 10 meters).
- **Working Principle:** PAN connects personal devices wirelessly (using Bluetooth, Infrared) or through USB cables, allowing devices to share data directly without needing a router or internet.

- **Example Technologies:** Bluetooth, USB connections, Infrared (IR), NFC (Near Field Communication).
- **Applications:** Connecting a smartphone to a wireless earphone (Bluetooth), connecting a wireless mouse/keyboard to a laptop, file sharing between two phones via Bluetooth.
- **Advantages:**
    1. Very easy to set up and use.
    2. No need for extra networking hardware like routers.
    3. Low power consumption (especially Bluetooth).
    4. Useful for quick, short-distance file transfer.
- **Limitations:**
    1. Very limited range (only a few meters).
    2. Low data transfer speed compared to LAN/WAN.
    3. Can only connect a limited number of devices at once.
    4. Prone to interference from other wireless signals.

1.3.3 LAN (LOCAL AREA NETWORK)

- **Definition:** A **Local Area Network (LAN)** is a privately-owned network that operates within a **single building, office, or campus**, typically covering a small geographical area.
- **Range:** Usually covers up to a few kilometers (like a single building or a college campus).
- **Working Principle:**
    - LAN connects computers, printers, and other devices using cables (Ethernet) or wireless (Wi-Fi) within a limited area.
    - LANs are widely used to share resources such as printers and exchange information within an organization.
    - LANs are distinguished by three main characteristics: **size**, **transmission technology**, and **topology**.
- **Speed:** LANs typically have data transfer speeds ranging from 10 Mbps to several Gbps (very fast compared to MAN/WAN).
- **Topologies used in LAN:** Bus, Star, Ring (explained in later units in detail, but LAN commonly uses **Star topology** with a central switch).
- **Standard/Protocol:** The most common LAN technology is **Ethernet (IEEE 802.3)**, and wireless LAN uses **Wi-Fi (IEEE 802.11)**.
- **Advantages:**
    1. High data transfer speed within a small area.
    2. Easy to set up, manage, and troubleshoot.
    3. Cost-effective since it covers a small area with limited hardware.
    4. Allows easy resource sharing (printers, files, internet connection).

- **Limitations:**
  1. Limited geographical coverage (cannot cover large areas like cities).
  2. Number of connected computers is limited.
  3. Security can be a concern if not properly configured.
  4. Requires cabling/infrastructure setup within the building.
- **Applications:** Office networks, college computer labs, home Wi-Fi networks, small business networks.

1.3.4 MAN (METROPOLITAN AREA NETWORK)

- **Definition:** A **Metropolitan Area Network (MAN)** covers a larger geographical area than LAN, typically spanning a **city or a large campus** connecting multiple LANs together.
- **Range:** Covers a range of about 5 to 50 kilometers (an entire city).
- **Working Principle:**
  - MAN connects multiple LANs (like different branch offices of the same company in a city) into a larger network.
  - It is often owned and operated by a single organization (like a cable TV company or a telephone company), but can also be a shared/public network.
  - MAN can support both **data and voice**, and can even carry cable television signals.
- **Best Known Example: Cable television network** available in many cities is a good example of MAN, and **WiMAX (IEEE 802.16)** is the standard technology used for wireless MAN.
- **Advantages:**
  1. Covers a larger area than LAN, connecting multiple offices/buildings in a city.
  2. Provides higher speed compared to WAN.
  3. Efficient for city-wide services like cable TV and metropolitan internet services.
  4. Reduces the cost of connecting multiple LANs individually to WAN.
- **Limitations:**
  1. More expensive to set up compared to LAN.
  2. Difficult to manage and maintain due to larger size.
  3. Security is harder to implement due to a bigger network.
  4. Requires more technical expertise to set up.
- **Applications:** City-wide cable TV networks, university campus networks spread across a city, government city networks.

1.3.5 WAN (Wide Area Network)

- **Definition:** A **Wide Area Network (WAN)** covers a very large geographical area, such as an **entire country or even multiple countries**, connecting different LANs and MANs together.
- **Range:** Can span across cities, countries, or even continents (hundreds to thousands of kilometers).
- **Working Principle:**
    - WAN connects multiple LANs/MANs located in different geographical locations using long-distance transmission media like **fiber optic cables, satellite links, or leased telephone lines**.
    - WAN typically consists of two main components:
        - **Hosts (End Systems):** These are the computers or devices used by end users, owned by individual customers/companies.
        - **Communication Subnet (or simply Subnet):** This carries messages from host to host and consists of **transmission lines** and **routers**.
    - **Transmission lines** move bits between machines (made of copper wire, optical fiber, or radio links).
    - **Routers** are specialized computers that connect two or more transmission lines and decide the best path to forward the data (this is called **routing**).
- **The Internet as WAN:** The **Internet** is the best example of a WAN, connecting millions of networks across the world.
- **Advantages:**
    1. Connects networks across very large geographical distances (even worldwide).
    2. Allows businesses to connect all their branch offices worldwide.
    3. Supports a huge number of users and devices globally.
    4. Enables global communication, business, and internet services.
- **Limitations:**
    1. Very expensive to set up and maintain due to long-distance infrastructure.
    2. Slower speed compared to LAN and MAN due to large distances.
    3. More prone to security threats and attacks due to its vast openness.
    4. Complex to troubleshoot and manage due to its huge scale.
- **Applications:** The Internet, banking networks connecting branches across countries, airline reservation systems, multinational company networks.

1.3.6 INTERNETWORKS

- **Definition:** When two or more different networks (like LAN, MAN, WAN) are connected together, the resulting combined network is called an **Internetwork**.
- **Purpose:** Internetworking allows devices on different types of networks (which may use different technologies) to communicate with each other seamlessly.
- **Device Used for Internetworking:** A **Gateway** is a special device used to connect two networks that use different protocols or technologies, allowing them to communicate.
- **Example:** The **Internet** itself is the largest example of an internetwork, connecting millions of LANs, MANs, and WANs across the world using common protocols (TCP/IP).
- **Working Principle:** When a device from one network wants to communicate with a device on a different network, the data passes through gateways/routers that translate and forward the data according to the destination network's protocol.
- **Applications:** The global Internet, connecting a company's LAN to the internet through an ISP (Internet Service Provider), connecting different branch office LANs through WAN links.

1.3.7 COMPARISON TABLE: LAN VS MAN VS WAN

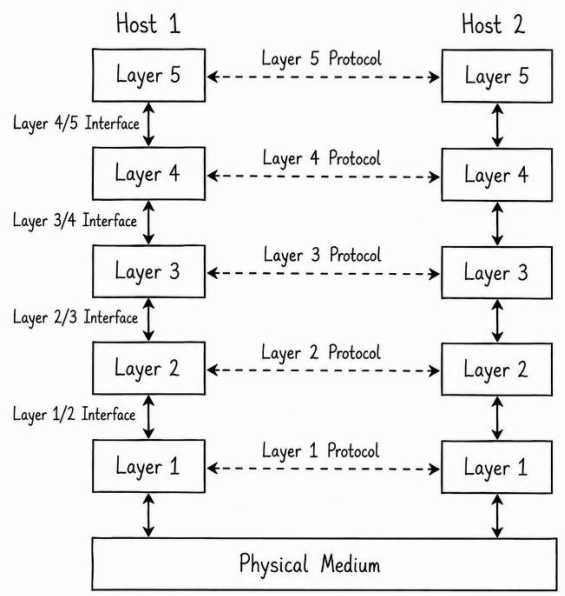
Comparison Point	LAN (Local Area Network)	MAN (Metropolitan Area Network)	WAN (Wide Area Network)
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Geographical Coverage	Small area (building, campus)	City-wide area	Country or worldwide
Ownership	Usually privately owned (single organization)	Owned by single or multiple organizations	Owned by multiple organizations/ISPs
Data Transfer Speed	Very high (10 Mbps to several Gbps)	Moderate (lower than LAN)	Comparatively lower due to distance
Setup Cost	Low cost	Medium cost	Very high cost
Maintenance	Easy to maintain	Moderate difficulty	Difficult to maintain
Transmission Medium	Twisted pair cables, Wi-Fi	Fiber optics, coaxial cables	Fiber optics, satellite links, leased lines
Example Technology	Ethernet, Wi-Fi	WiMAX, Cable TV network	Internet, MPLS
Example	College Lab, Office network	City cable TV network	The Internet, banking networks

1.4 NETWORK SOFTWARE

1.4.1 PROTOCOL HIERARCHIES

- **Definition of Protocol:** A **protocol** is an agreement between the communicating parties on how communication should take place; it defines a set of rules that both sender and receiver must follow.
- **Need for Protocol Hierarchies (Layers):**
  - Network communication is a very complex task, so instead of writing one large complex program to handle everything, the task is divided into smaller, manageable parts called **layers**.
  - Each layer is responsible for performing a specific function, making the design, implementation, and troubleshooting of networks much easier.
  - This concept of organizing protocols into layers is called a **Protocol Hierarchy** or **Protocol Stack**.
- **How Protocol Hierarchy Works:**
  - **Layer n on one machine communicates with Layer n on another machine** — this is called a **peer-to-peer communication** at that layer, and the rules used for this communication are called the **protocol of layer n**.
  - In reality, no data is directly transferred from layer n on one machine to layer n on another machine (except at the lowest physical layer). Instead, each layer passes data and control information to the layer immediately below it.
  - Between each pair of adjacent layers, there is an **interface**, which defines the primitive operations and services that the lower layer offers to the layer above it.
  - A set of layers and protocols is called **network architecture**.

Text-Based Diagram: Protocol Hierarchy (Layered Communication)



- **1.4.1.5 Important Terms in Protocol Hierarchy**
  - **Entities:** The active elements in each layer are called entities. They can be software (like a process) or hardware (like an intelligent I/O chip).
  - **Peer:** Entities on the same layer on different machines are called **peers** (e.g., layer 3 on machine A and layer 3 on machine B are peers).
  - **Service:** A set of operations that a layer provides to the layer above it.
  - **Service Provider / Service User:** The lower layer is the service provider, and the layer above it is the service user.

1.4.2 DESIGN ISSUES FOR LAYERS

- **Purpose:** These are the important problems/questions that must be solved while designing every layer of a network architecture.
- **Addressing**
  - Every layer needs a mechanism to identify senders and receivers, because a network may have many computers and processes, and a message must reach the correct destination.
- **Error Control**
  - Since physical communication is never perfect, layers must have agreed-upon **error-detecting and error-correcting** codes so that receivers can identify and possibly fix errors in the transmitted data.
- **Flow Control**
  - If a fast sender sends data faster than a slow receiver can handle, it will overload the receiver, so a mechanism is needed to control the sending rate (feedback from receiver to sender).
- **Multiplexing and Demultiplexing**
  - It may be inefficient or costly to set up a separate connection for each pair of communicating processes, so layers need mechanisms to allow **multiplexing** (combining multiple communications onto a single connection) and **demultiplexing** (separating them again at the destination).
- **Routing**
  - When there are multiple paths between source and destination, a decision must be made on **which route/path** the data should take — this is especially important when there are multiple routers/networks involved (i.e., not just a direct connection).

1.4.3 CONNECTION-ORIENTED AND CONNECTION-LESS SERVICES

1. **Connection-Oriented Service**
   - **Definition:** In connection-oriented service, a **connection is first established** between sender and receiver before any actual data is transferred, similar to how a telephone system works.
   - **Working Principle:**
     - The sender and receiver first perform a **"handshake"** to set up the connection.
     - Once established, data is sent as a **continuous stream of bytes/packets** in the correct order.
     - After all data is transferred, the connection is properly **released (terminated)**.
   - **Real-World Analogy:** Making a phone call — you first dial and connect to the person, talk (send data), and then hang up (release connection).
   - **Example Protocol: TCP (Transmission Control Protocol)** is a connection-oriented protocol.
   - **Advantages:**
     1. Reliable data delivery since connection is confirmed before sending data.
     2. Data arrives in the correct order (sequenced) at the receiver.
     3. Suitable for applications requiring accuracy, like file transfer.
     4. Errors can be detected and corrected due to acknowledgment mechanisms.
   - **Limitations:**
     1. Takes extra time to establish a connection before actual data transfer.
     2. More overhead due to handshaking and acknowledgment messages.
     3. Not suitable for very fast, real-time applications (like live video where slight delay is acceptable but connection setup delay is not).
     4. Consumes more network resources due to maintaining connection state.

2. **Connection-Less Service**
   - **Definition:** In connection-less service, **no prior connection is established**; data (in the form of independent packets, called **datagrams**) is sent directly to the destination without any handshake.
   - **Working Principle:**
     - Each packet (datagram) is sent independently and **may take a different route** to reach the destination.
     - Packets **may not arrive in the same order** they were sent, since each one is routed independently.

    - There is **no guarantee** of delivery — packets can be lost, duplicated, or delayed.
- **Real-World Analogy:** Sending postal letters — each letter is posted independently, may take different routes/times, and there's no guarantee all will arrive or arrive in order.
- **Example Protocol: UDP (User Datagram Protocol)** is a connection-less protocol.
- **Advantages:**
    1. Faster transmission since no time is wasted in connection setup.
    2. Less overhead as no acknowledgment or handshaking is required.
    3. Suitable for real-time applications like video streaming, online gaming, and voice calls.
    4. Simple to implement compared to connection-oriented service.
- **Limitations:**
    1. No guarantee of reliable delivery (packets may be lost).
    2. Packets may arrive out of order at the receiver.
    3. No flow control or congestion control mechanism by default.
    4. Not suitable for applications requiring 100% data accuracy (like file transfer, bank transactions).

Comparison Table: Connection-Oriented vs Connection-Less Service

Comparison Point	Connection-Oriented Service	Connection-Less Service
Connection Setup	Connection must be established before data transfer	No connection setup required
Analogy	Similar to telephone system	Similar to postal system
Reliability	Highly reliable	Not reliable (best-effort delivery)
Order of Delivery	Data arrives in the same order sent	Data may arrive out of order
Speed	Slower due to connection setup overhead	Faster since no setup delay
Example Protocol	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Acknowledgment	Uses acknowledgment for every packet	No acknowledgment required
Overhead	High overhead (handshaking, ACKs)	Low overhead
Suitable Applications	File transfer, email, web browsing	Video streaming, online gaming, VoIP
Error/Flow Control	Present (built-in)	Usually absent (handled by application if needed)

1.4.4 SERVICE PRIMITIVES

- **Definition:** A **service primitive** (or simply **primitive**) is a specific action or operation that a service allows the user to perform in order to access that service; primitives tell the service to perform some action or to report on an action taken by a peer entity.
- **Purpose:** Service primitives allow the layer above (service user) to request specific services from the layer below (service provider), such as establishing a connection or sending data.
- **Five Basic Service Primitives (used for connection-oriented service):**

Primitive	Meaning
LISTEN	The server is ready and waiting for an incoming connection request.
CONNECT	The client initiates and establishes a connection with the server.
RECEIVE	Used to block the server, waiting for an incoming connection request.
SEND	Used to send data/messages to the receiver over the established connection.
DISCONNECT	Used to terminate/release the connection after communication is complete.

- **Working Example (Client-Server Communication using Primitives):**
  - Step 1: Server executes **LISTEN** to indicate it is ready to accept a connection.
  - Step 2: Server executes **RECEIVE** and blocks (waits) until a client connects.
  - Step 3: Client executes **CONNECT** to establish a connection with the server.
  - Step 4: Once connected, both client and server can use **SEND** to exchange data.
  - Step 5: After the communication is finished, either side executes **DISCONNECT** to terminate the connection.
  - **Exam Tip:** Draw a simple sequence diagram showing Client and Server with arrows labeled LISTEN → CONNECT → SEND → DISCONNECT, this is commonly asked in a 4-5 mark question.

1.4.5 RELATIONSHIP BETWEEN SERVICES AND PROTOCOLS

- **Definition of Service:** A service is a **set of operations (primitives)** that a layer provides to the layer above it; a service defines **what operations the layer can perform** for its users, without specifying how these operations are implemented.
- **Definition of Protocol:** A protocol is a **set of rules** governing the format and meaning of the packets/messages that are exchanged by peer entities within a layer; a protocol defines **how a service is actually implemented**.
- **Key Difference (Very Important Concept):**
  - A **service** relates to the interface between two layers (the layer below and the layer above), whereas a **protocol** relates to the packets sent between peer entities on different machines (same layer, different machine).

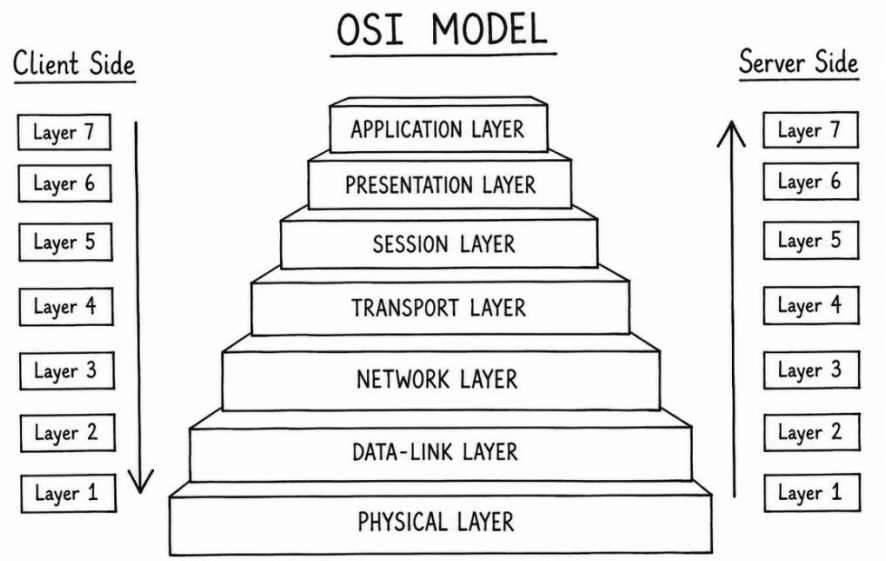
  - Services are like **abstract functions/interfaces** available to the upper layer, while protocols are the **actual implementation rules** used to provide those services.
  - **Analogy:** A service is like a company promising to deliver "next day parcel delivery" (what is offered), while a protocol is like the actual internal process/rules the company follows to achieve this delivery (how it is done, e.g., truck routes, sorting process).
- **Why This Distinction Is Important:**
  - A service only tells **what** a layer does for the layer above it, and completely hides the details of how the layer implements it.
  - Protocols can be **changed/upgraded** without affecting the layers above, as long as the service provided to the upper layer remains the same — this makes network design flexible and modular.
- **Exam Tip:** "Differentiate between Service and Protocol" is a classic 4-5 mark or comparison-based question — remember the analogy (service = what is offered, protocol = how it is implemented) to answer confidently.

1.5 REFERENCE MODELS

1.5.1 OSI REFERENCE MODEL (OPEN SYSTEMS INTERCONNECTION)

- **Definition:** The **OSI Model** is a conceptual framework developed by **ISO (International Organization for Standardization)** that divides network communication into **7 layers**, where each layer has specific responsibilities for data communication.
- **Purpose of the OSI Model**
  - To provide a **standard reference structure** so that different vendors' hardware/software can communicate with each other regardless of underlying technology.
  - To divide the complex task of networking into smaller, manageable, and independent layers.
- **Need for Layering (Why Layered Approach is Used)**
  - Each layer performs a **specific, well-defined task**, which makes the system's design simple, organized, and easy to understand.
  - Layers allow changes to be made in one layer **without affecting** other layers (as long as the interface between layers remains the same).
  - Layering makes **troubleshooting easier**, since problems can be isolated to a specific layer.
  - Layering promotes **standardization**, allowing different manufacturers to design compatible networking equipment.

- **Text-Based Diagram: OSI Model (7 Layers)**



- **How to Draw in Exam:** Draw 7 rectangular boxes stacked vertically, with **Layer 7 (Application) at the top** and **Layer 1 (Physical) at the bottom**. Label each box clearly with the layer number and name. This is one of the **most important diagrams** in the entire subject — practice drawing it quickly and neatly.
- **Memory Technique (Mnemonic) to Remember OSI Layers**
  - From **Layer 7 to Layer 1** (top to bottom): **"All People Seem To Need Data Processing"**
    - **A**pplication, **P**resentation, **S**ession, **T**ransport, **N**etwork, **D**ata Link, **P**hysical.
  - From **Layer 1 to Layer 7** (bottom to top): **"Please Do Not Throw Sausage Pizza Away"**
    - **P**hysical, **D**ata Link, **N**etwork, **T**ransport, **S**ession, **P**resentation, **A**pplication.
  - **Exam Tip:** Write this mnemonic in the margin of your answer sheet as a quick reference while attempting the question.
- **Functions and Responsibilities of Each Layer (Detailed)**
  - **Layer 1 – Physical Layer:**
    - **Function:** Deals with transmission of raw **bits (0s and 1s)** over a physical communication channel/medium.
    - **Responsibilities:** Defines the electrical, mechanical, and timing interface to the transmission medium; deals with voltage levels, cable specifications, connectors, and data rates.
    - **Concerned with:** How many volts represent a 0 or 1, how many pins the connector has, transmission medium type (copper, fiber, wireless).

- **Protocols/Standards:** Ethernet (physical specifications), RS-232, USB physical layer.

- **Layer 2 – Data Link Layer:**
  - **Function:** Responsible for reliable transfer of data (organized into **frames**) between two directly connected nodes over the physical layer.
  - **Responsibilities: Framing** (dividing data into frames), **error detection and correction**, **flow control**, and **access control** (deciding which device gets to use the shared medium, especially in broadcast networks).
  - **Sub-layers:** Divided into **MAC (Media Access Control)** and **LLC (Logical Link Control)** sub-layers.
  - **Protocols/Standards:** Ethernet (IEEE 802.3), PPP (Point-to-Point Protocol), HDLC.
- **Layer 3 – Network Layer:**
  - **Function:** Responsible for **logical addressing** and **routing** of data packets from source to destination across multiple networks.
  - **Responsibilities:** Determines the **best path (route)** for data to travel from source to destination, handles **congestion control**, and manages logical addressing (**IP addresses**).
  - **Protocols/Standards:** IP (Internet Protocol), ICMP, ARP, RIP, OSPF.
- **Layer 4 – Transport Layer:**
  - **Function:** Provides **end-to-end (process-to-process) delivery** of the entire message, ensuring data reaches the correct application/process on the destination machine.
  - **Responsibilities: Segmentation** of data, **flow control**, **error control**, and **connection management** (establishing, maintaining, terminating connections); also performs multiplexing/demultiplexing.
  - **Protocols/Standards:** TCP (Transmission Control Protocol), UDP (User Datagram Protocol).
- **Layer 5 – Session Layer:**
  - **Function:** Responsible for establishing, managing, and terminating **sessions** (a session is a continuous exchange of information, like a login session) between two communicating applications.
  - **Responsibilities: Dialog control** (deciding whose turn it is to transmit — half-duplex/full-duplex), **synchronization** (inserting checkpoints in data so that if a crash occurs, only data after the last checkpoint needs to be resent).
  - **Real-World Example:** Logging into a bank website and maintaining that logged-in session while browsing multiple pages.

- **Layer 6 – Presentation Layer:**
  - **Function:** Concerned with the **syntax and semantics** of the data being transmitted; ensures data is presented in a format understandable by the receiving application.
  - **Responsibilities: Data translation/conversion** (between different data formats used by different systems), **data compression** (to reduce size of data before transmission), and **data encryption/decryption** (for security purposes).
  - **Real-World Example:** Converting data into a standard format like ASCII/Unicode, or compressing a file before sending it over email.
- **Layer 7 – Application Layer:**
  - **Function:** The topmost layer that provides **network services directly to the end-user applications**, acting as the interface between the user and the network.
  - **Responsibilities:** Supports application services like **email, file transfer, web browsing, and remote login**.
  - **Protocols/Standards:** HTTP, FTP, SMTP, DNS, Telnet.

- **Data Encapsulation Process (How Data Moves Down the Layers at Sender Side)**
  - As data moves down from Application Layer to Physical Layer, **each layer adds its own header (and sometimes trailer)** information to the data received from the layer above it — this process is called **Encapsulation**.
  - **Step-by-Step Encapsulation:**
    1. **Application Layer:** User data is created (e.g., an email message).
    2. **Presentation Layer:** Adds formatting/encryption info, converts data into a standard format.
    3. **Session Layer:** Adds session-related control information for managing the dialog.
    4. **Transport Layer:** Breaks data into **segments** and adds a **Transport Header** (contains port numbers, sequence numbers). Now called a **Segment**.
    5. **Network Layer:** Adds a **Network Header** (contains source and destination IP addresses). Now called a **Packet**.
    6. **Data Link Layer:** Adds a **Data Link Header and Trailer** (contains MAC addresses and error-checking information). Now called a **Frame**.
    7. **Physical Layer:** Converts the frame into raw **bits (0s and 1s)** for actual transmission over the medium.

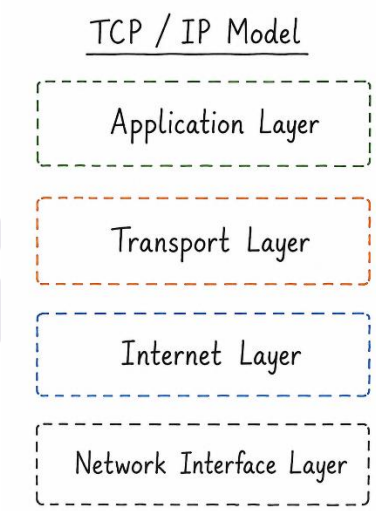
- **Data Decapsulation Process (How Data Moves Up the Layers at Receiver Side)**
  - At the receiver side, the exact **reverse process** happens — each layer **removes (strips off)** its corresponding header/trailer as data moves up from Physical Layer to Application Layer, and passes the remaining data to the layer above.
  - **Step-by-Step Decapsulation:**
    1. **Physical Layer:** Receives raw bits and converts them into a frame.
    2. **Data Link Layer:** Removes the Data Link header/trailer, checks for errors, and passes the packet to the Network Layer.
    3. **Network Layer:** Removes the Network header, checks the destination IP address, and passes the segment to the Transport Layer.
    4. **Transport Layer:** Removes the Transport header, reassembles segments, and passes the data to the Session Layer.
    5. **Session Layer:** Removes session control information and passes data to the Presentation Layer.
    6. **Presentation Layer:** Converts data back into a readable format (decryption/decompression) and passes it to the Application Layer.
    7. **Application Layer:** The final data is delivered to the user application (e.g., the email is displayed to the user).

- **Advantages of OSI Model:**
  1. Provides a clear, standardized, and universally accepted reference for network communication.
  2. Layers can be developed, modified, or replaced independently without affecting other layers.
  3. Makes troubleshooting easier since problems can be isolated to a specific layer.
  4. Helps in the design of interoperable network devices from different manufacturers.

- **Limitations of OSI Model:**
  1. It is a theoretical/conceptual model; it was designed before the protocols were developed, so it is not directly implemented in real-world networks.
  2. Some layers (like Session and Presentation) are rarely used as separate layers in real implementations; their functions are often merged into other layers.
  3. It is more complex compared to the TCP/IP model due to having 7 layers.
  4. Fitting existing protocols (like TCP/IP protocols) into the OSI model can sometimes be difficult and confusing.

1.5.2 TCP/IP REFERENCE MODEL

- **Definition:** The **TCP/IP Model** is a practical, real-world networking model developed by the **US Department of Defense (DoD)**, consisting of **4 layers**, and it is the actual model used to build the modern Internet.
- **Purpose of the TCP/IP Model**
  - To create a network that could **survive the loss of hardware** (like during war), ensuring existing communication could continue even if some machines/routers went down.
  - To design a flexible architecture that could handle **multiple types of applications** and adapt to a wide variety of underlying network technologies.
- **Text-Based Diagram: TCP/IP Model (4 Layers)**



- **How to Draw in Exam:** Draw 4 stacked boxes, with Application Layer on top and Host-to-Network Layer at the bottom. This diagram is simpler than OSI, so make sure to draw it quickly and clearly.
- **Functions and Responsibilities of Each Layer (Detailed)**
  - **Layer 1 – Host-to-Network Layer (Link Layer):**
    - **Function:** This layer is not clearly defined in the original TCP/IP model; it is responsible for the actual physical connection of the host to the network, combining the functions of OSI's Physical and Data Link layers.
    - **Responsibilities:** Handles the physical transmission of data, framing, and hardware addressing.
  - **Layer 2 – Internet Layer:**
    - **Function:** Responsible for allowing hosts to send **packets** into any network and have them travel independently to the destination (possibly through different routers), similar to the OSI Network Layer.

    - **Responsibilities: Logical addressing (IP addressing)**, **routing** of packets, and handling **congestion control**.
    - **Main Protocol: IP (Internet Protocol)** is the key protocol of this layer; also includes ICMP and ARP.
  - **Layer 3 – Transport Layer:**
    - **Function:** Designed to allow **peer entities on source and destination hosts to have a conversation**, similar to the OSI Transport Layer.
    - **Responsibilities:** Provides **end-to-end data delivery**, error control, and flow control.
    - **Main Protocols: TCP** (connection-oriented, reliable) and **UDP** (connection-less, fast but unreliable).
  - **Layer 4 – Application Layer:**
    - **Function:** This layer combines the functions of OSI's **Session, Presentation, and Application layers** into a single layer, providing all high-level protocols directly to user applications.
    - **Responsibilities:** Provides protocols for virtual terminal, file transfer, email, and other user-level services.
    - **Main Protocols:** HTTP, FTP, SMTP, DNS, Telnet.

- **Data Encapsulation in TCP/IP Model**

Similar to OSI, TCP/IP model also encapsulates data as it moves down the layers:

1. **Application Layer:** Creates user data (Data).
2. **Transport Layer:** Adds TCP/UDP header → forms a **Segment**.
3. **Internet Layer:** Adds IP header → forms a **Packet (Datagram)**.
4. **Host-to-Network Layer:** Adds header/trailer for physical transmission → forms a **Frame**, then converts to **bits**.

- **Advantages of TCP/IP Model:**
  1. It is a practical model, and it is the actual model used to run the real Internet worldwide.
  2. It is simpler with only 4 layers, making it easier to understand and implement compared to OSI.
  3. It is an open standard, not owned by any single company, so it can be used and implemented by anyone.
  4. It is highly scalable and flexible, supporting a large variety of routing protocols and applications.

- **Limitations of TCP/IP Model:**
  1. It does not clearly distinguish between service, interface, and protocol concepts (unlike OSI, which is very clear about this).

2. The Host-to-Network layer is not a true layer in the technical sense, as it is not clearly defined.
3. It is not a general/universal model, since it was designed to fit the specific protocols it was built around (like TCP and IP).

1.5.3 COMPARISON TABLE: OSI MODEL VS TCP/IP MODEL

Comparison Point	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Developed By	ISO (International Organization for Standardization)	US Department of Defense (DoD)
Number of Layers	7 layers	4 layers
Approach	Theoretical/conceptual model, designed first, then protocols were fitted	Practical model, developed based on existing protocols
Layer Names	Physical, Data Link, Network, Transport, Session, Presentation, Application	Host-to-Network, Internet, Transport, Application
Usage	Used as a reference/teaching model, rarely implemented directly	Actually implemented and used to run the real Internet
Reliability of Service	Both connection-oriented and connection-less supported at Network layer	Only connection-less supported at Internet layer (connection-oriented only at Transport layer)
Protocol Dependency	Protocol-independent (designed before protocols)	Protocol-dependent (designed around existing TCP/IP protocols)
Session & Presentation Layers	Present as separate, dedicated layers	Not present separately; merged into Application layer
Flexibility	Less flexible for real-world implementation due to complexity	More flexible and adaptable, hence widely adopted
Popularity	Used mainly for understanding/teaching networking concepts	Used practically in real-world networks (the actual Internet)

1.6 PHYSICAL LAYER

- **Definition and Purpose:** The **Physical Layer** is the lowest layer of the OSI model, responsible for the actual transmission of raw **bits (0s and 1s)** over a physical communication medium between devices.
- **Why This Layer is Important:** Without the physical layer, no actual electrical/optical/wireless signal transmission would be possible — it forms the actual "road" on which data travels.
- **Two Broad Categories of Transmission Media (as per SPPU Syllabus):**
  - **Guided Transmission Media** (wired/physical medium).
  - **Wireless Transmission** (unguided medium, through air/space).

1.6.4 GUIDED TRANSMISSION MEDIA

Definition: Guided media are those transmission media in which signals are **guided/directed along a solid physical path/medium**, such as a cable.

1. **Twisted Pair Cable**
   - **Structure:** Consists of **two insulated copper wires twisted together** in a helical (spiral) shape; twisting reduces electrical interference from nearby pairs (called **crosstalk**).
   - **Types:**
     - **UTP (Unshielded Twisted Pair):** No extra shielding, cheaper, commonly used in telephone lines and LANs.
     - **STP (Shielded Twisted Pair):** Has extra metallic shielding around the wires to reduce interference, more expensive, used in environments with high electrical noise.
   - **Applications:** Telephone networks, Ethernet LAN cables (Cat5, Cat6 cables).
   - **Advantages:**
     1. Low cost and easy to install.
     2. Flexible and lightweight cable.
     3. Widely available and easy to maintain.
     4. Suitable for short to medium distance communication.
   - **Limitations:**
     1. Susceptible to electromagnetic interference (EMI) and crosstalk.
     2. Limited bandwidth and lower data transmission speed compared to fiber optics.
     3. Signal attenuation (weakening) occurs over long distances.
     4. Less secure (data can be tapped easily compared to fiber).

2. **Coaxial Cable**
   - **Structure:** Consists of a **central copper conductor**, surrounded by an **insulating layer**, then a **metallic shield (braided mesh)**, and an outer **protective plastic covering**.
   - **Types:** Baseband coaxial cable (used for digital transmission) and Broadband coaxial cable (used for analog transmission, like cable TV).
   - **Applications:** Cable television networks, older Ethernet networks, long-distance telephone lines.
   - **Advantages:**
     1. Better shielding than twisted pair, so less interference.
     2. Higher bandwidth compared to twisted pair cable.
     3. Can carry signals over longer distances than twisted pair.
     4. More durable and reliable than twisted pair.
   - **Limitations:**
     1. More expensive than twisted pair cable.
     2. Bulkier and less flexible, harder to install.
     3. Still has more signal loss compared to fiber optic cables.
     4. Being replaced by fiber optics in most modern high-speed networks.

3. **Fiber Optic Cable**
   - **Structure:** Consists of a **glass or plastic core** that carries light signals, surrounded by a **cladding layer** (which reflects light back into the core), and an outer protective jacket.
   - **Working Principle:** Data is transmitted as **pulses of light** through the glass/plastic core, using the principle of **total internal reflection** to keep the light signal bouncing within the core without escaping.
   - **Types: Single-mode fiber** (thin core, long-distance, uses laser light) and **Multi-mode fiber** (thicker core, shorter distance, uses LED light).
   - **Applications:** High-speed internet backbones, long-distance telecommunication networks, submarine communication cables.
   - **Advantages:**
     1. Extremely high bandwidth and very high-speed data transmission.
     2. Immune to electromagnetic interference (since it uses light, not electricity).
     3. Very low signal loss (attenuation), suitable for very long distances.
     4. More secure since it is very difficult to tap fiber optic cables without detection.

- **Limitations:**
    1. Very expensive to install and maintain compared to copper cables.
    2. Requires specialized skill and equipment for installation and splicing.
    3. Fragile and can break easily if bent too sharply.
    4. Connecting/joining fiber cables (splicing) is a complex and costly process.

1.6.4.4 Comparison Table: Twisted Pair vs Coaxial vs Fiber Optic Cable

Comparison Point	s	Coaxial Cable	Fiber Optic Cable
Medium Type	Copper wires (twisted)	Copper (central conductor + shield)	Glass/plastic core
Signal Type	Electrical signals	Electrical signals	Light signals
Bandwidth	Low	Medium	Very High
Cost	Cheapest	Medium cost	Most expensive
Interference	High (prone to EMI/crosstalk)	Medium (shielded, less interference)	None (immune to EMI)
Distance Coverage	Short distance	Medium distance	Very long distance
Installation	Easy	Moderate	Difficult (needs expertise)
Security	Low (easy to tap)	Medium	Very high (hard to tap)
Speed	Low to moderate	Moderate	Very high speed
Applications	Telephone lines, LAN	Cable TV, older LAN	Internet backbone, long-distance links

1.6.5 WIRELESS TRANSMISSION

Definition: Wireless transmission (unguided media) refers to the transfer of data through **air or space** using electromagnetic waves, without any physical cable/wire.

1. **Electromagnetic Spectrum (Basis of Wireless Transmission)**
    - Wireless communication uses different parts of the **electromagnetic spectrum**, including radio waves, microwaves, infrared, and light waves, depending on the frequency used.
    - Higher frequency generally allows higher data rates but travels shorter distances and is affected more by obstacles.

2. **Radio Transmission**
   - **Working Principle:** Radio waves are easy to generate, can travel long distances, and can penetrate through walls easily, making them useful for both indoor and outdoor communication.
   - **Characteristics:** Radio waves are **omnidirectional**, meaning they travel in all directions from the source, so the sender and receiver do not need to be precisely aligned.
   - **Applications:** AM/FM radio broadcasting, walkie-talkies, wireless LAN (Wi-Fi in some cases), cordless phones.
3. **Microwave Transmission**
   - **Working Principle:** Above 100 MHz, waves travel in nearly straight lines and can be narrowly focused, requiring the **sending and receiving antennas to be precisely aligned** with each other (line-of-sight communication).
   - **Characteristics:** Since microwaves travel in a straight line, curvature of the earth limits transmission distance; tall towers are used to extend the range, and repeater stations are needed for very long distances.
   - **Applications:** Long-distance telephone communication, satellite communication, point-to-point microwave links between buildings.
4. **Infrared Transmission**
   - **Working Principle:** Infrared signals are used for **short-range wireless communication** and cannot pass through solid objects/walls (which is actually an advantage for security, preventing interference between adjacent rooms).
   - **Characteristics:** Requires **line-of-sight** between transmitter and receiver; relatively low cost and easy to build.
   - **Applications:** TV remote controls, short-range device communication (older mobile phones, some IoT devices).

Comparison Table: Radio vs Microwave vs Infrared Transmission

Comparison Point	Radio Waves	Microwave	Infrared
Frequency Range	Low frequency	High frequency	Very high frequency (near light)
Directionality	Omnidirectional	Directional (line-of-sight)	Directional (line-of-sight)
Obstacle Penetration	Can pass through walls	Cannot pass through solid obstacles	Cannot pass through solid objects
Range	Long range	Long range (with towers/repeaters)	Short range

Alignment Needed	Not required	Precise alignment required	Precise alignment required
Applications	Radio broadcast, Wi-Fi	Satellite, long-distance telephony	TV remotes, short-range devices

Comparison Table: Guided Media vs Unguided Media

Comparison Point	Guided Media	Unguided Media (Wireless)
Definition	Signal travels through a physical solid medium	Signal travels through air/space
Examples	Twisted pair, coaxial, fiber optic cable	Radio waves, microwaves, infrared
Physical Connection	Requires physical cable connection	No physical cable required
Mobility	Limited (devices must stay connected to cable)	High mobility (devices can move freely)
Interference	Less affected by environmental factors	More affected by weather, obstacles, interference
Security	Relatively more secure (needs physical access to tap)	Less secure (signals can be intercepted from air)
Installation Cost	Higher initial installation cost (cabling)	Lower installation cost (no cabling needed)
Distance Limitation	Distance limited by cable length/attenuation	Can cover very large areas (with towers/satellites)
Example Use Case	Wired office LAN, telephone lines	Mobile networks, Wi-Fi, satellite communication

1.6.6 TELEPHONE SYSTEM

- **Purpose:** The **telephone system** was originally designed for **voice communication**, but it has become one of the most important infrastructures used for **data communication** (like dial-up internet, DSL) as well.
- **Structure of Telephone System:**
    - **Local Loop:** The wire connecting a subscriber's telephone to the nearest telephone exchange (usually twisted pair cable).
    - **Trunks:** High-capacity communication links that connect different telephone exchanges/switching offices together (usually fiber optic or microwave links).
    - **Switching Offices (Exchanges):** Devices that connect calls by switching/routing them between local loops and trunks.

- **Working Principle:**
  - When a person makes a call, the voice signal travels through the **local loop** to the nearest **switching office**.
  - The switching office finds a path (through trunks and possibly other switching offices) to reach the destination telephone's local loop.
  - Modern telephone systems use **digital switching** internally, even though the local loop to the customer's phone is often analog.
- **Use of Telephone System for Data Communication:**
  - **Modems** are used to convert digital computer data into analog signals (suitable for transmission over telephone lines) and vice versa — this process is called **Modulation and Demodulation**.
  - Technologies like **DSL (Digital Subscriber Line)** use the existing telephone line infrastructure to provide high-speed internet access to homes.
- **Applications:** Landline voice calls, DSL internet connections, fax machines, older dial-up internet connections.

1.6.7 NARROWBAND AND BROADBAND COMMUNICATION SYSTEMS

1. **Narrowband Communication System**
   - **Definition: Narrowband** refers to a communication system that uses a **small range of frequencies (narrow bandwidth)** to transmit data, resulting in **lower data transmission speed**.
   - **Characteristics:** Suitable for transmitting simple data like voice calls or basic text messages, but not suitable for high-speed data like video streaming.
   - **Applications:** Traditional telephone voice calls, older dial-up internet connections, walkie-talkies.
   - **Advantages:**
     1. Requires less bandwidth, so more channels can fit in a given frequency range.
     2. Signal can travel longer distances with less power (compared to broadband in some cases).
     3. Simple technology, cost-effective for basic communication needs.
     4. Less susceptible to certain types of interference due to narrow frequency use.
   - **Limitations:**
     1. Very low data transmission speed, unsuitable for modern high-speed applications.
     2. Cannot support multiple simultaneous services on the same line efficiently.
     3. Not suitable for video streaming, large file transfers, or high-quality multimedia applications.
     4. Becoming outdated with the rise of high-speed broadband technologies.

2. **Broadband Communication System**
   - **Definition: Broadband** refers to a communication system that uses a **wide range of frequencies (wide bandwidth)** to transmit large amounts of data at very **high speed**.
   - **Characteristics:** Multiple signals/channels can be transmitted simultaneously over the same medium using **different frequency bands** (this technique is called **Frequency Division Multiplexing**), allowing much higher overall data capacity.
   - **Applications:** High-speed home internet (fiber broadband, cable broadband), video streaming services (Netflix, YouTube), video conferencing.
   - **Advantages:**
     1. Provides very high data transmission speed, suitable for modern applications.
     2. Supports multiple services simultaneously (like internet, voice, and TV) over the same connection.
     3. Suitable for high-quality video streaming, gaming, and large file downloads.
     4. Always-on connection (unlike old dial-up, which required dialing each time).
   - **Limitations:**
     1. More expensive infrastructure setup compared to narrowband.
     2. Requires more complex equipment (modems, routers) for proper functioning.
     3. May not be available in all remote/rural areas due to infrastructure limitations.
     4. Higher power consumption in some broadband equipment compared to simple narrowband devices.

1.6.7.3 Comparison Table: Narrowband vs Broadband Communication System

Comparison Point	Narrowband	Broadband
Bandwidth	Narrow/small frequency range	Wide/large frequency range
Data Speed	Low data transmission speed	High data transmission speed
Number of Channels	Fewer channels/services at a time	Multiple channels/services simultaneously
Cost	Lower cost	Higher cost
Technology Used	Simple, older technology	Advanced, modern technology
Applications	Voice calls, basic text/dial-up internet	Video streaming, high-speed internet, video calls
Multiplexing	Limited or no multiplexing	Uses techniques like Frequency Division Multiplexing
Connection Type	Often requires dial-up (not always-on)	Always-on connection

Suitability for Multimedia	Not suitable	Highly suitable
Present-Day Relevance	Becoming outdated	Widely used and preferred today

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IMPORTANT SPPU EXAM QUESTIONS – UNIT I

1. Define computer network. Explain the basic components required for network communication.
2. Explain the uses of computer networks in business applications, home applications, and for mobile users.
3. What is network hardware? Explain PAN, LAN, MAN, and WAN with suitable examples.
4. Compare LAN, MAN, and WAN based on at least 8 parameters.
5. What is an internetwork? Explain the role of a gateway in internetworking.
6. Explain broadcast networks and point-to-point networks with examples.
7. What is a protocol hierarchy? Explain with a neat diagram how communication takes place between layers.
8. Explain the design issues for layers in computer networks.
9. Differentiate between connection-oriented and connection-less services with examples.
10. What are service primitives? Explain the five basic service primitives used in connection-oriented communication.
11. Explain the relationship between services and protocols with a suitable example/analogy.
12. Explain the OSI reference model in detail with a neat diagram. Describe the function of each layer.
13. Explain the TCP/IP reference model in detail with a neat diagram. Describe the function of each layer.
14. Compare the OSI model and the TCP/IP model based on at least 8-10 parameters.
15. Explain the data encapsulation and decapsulation process in the OSI model with a diagram.
16. Explain guided transmission media in detail. Compare twisted pair, coaxial cable, and fiber optic cable.
17. Explain wireless transmission. Differentiate between radio waves, microwave, and infrared transmission.
18. Explain the working of the telephone system, including local loop, trunks, and switching offices.
19. Differentiate between narrowband and broadband communication systems with examples.
20. Write a short note on the case study: Comparison of OSI vs TCP/IP in real-world networks.

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State and explain the ACID properties. During its execution a transaction passes through several states until it finally commits or aborts. List all possible sequences of states. Explain when each state transition occurs. (Appeared 5/8 papers — exact same question statement every time)

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